HWW3: multi-resolution HEALPix unstructured grids for WAVEWATCH III

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Internal note, draft of November 13, 2026

Abstract

We build unstructured WAVEWATCH III (WW3) grids whose nodes are the centres of HEALPix cells of different resolutions: fine near the coast, coarse in the open ocean. The model code is unchanged. Because every node is a NESTED HEALPix pixel, the model state is a multi-resolution HEALPix map. It can be stored as GRID4EARTH Zarr datasets and compared with satellite products on the same grid without interpolation. This note describes the grid construction, how it is validated, and a first North Atlantic simulation for January 2026. [TODO: results]

1 Motivation

Unstructured wave models usually rely on meshes from generic mesh generators. Their nodes do not belong to any global indexing system, so every comparison with observations or with another model needs an interpolation. HEALPix offers a hierarchical, equal-area and iso-latitude tessellation of the sphere. Each cell at N_{side} splits into four cells at $2N_{\mathrm{side}}$, so multi-resolution grids come from simple quadtree refinement, and each node has a global integer identifier (the NESTED *uniq* index). Placing the WW3 nodes at HEALPix cell centres keeps the flexibility of unstructured grids while making the outputs native to discrete global grid systems.

2 Grid construction

2.1 Refinement

We start from the cells at $N_{\mathrm{side}}^{\mathrm{min}}$ whose centres lie in the domain, expanded by a margin. A cell of angular radius r is split into its four children when its lower-bound distance to the coastline, $d - r$, is below the distance D_k of a refinement rule k whose target resolution $N_{\mathrm{side}}^{(k)}$ is finer than the cell. The coastline comes from the GSHHS land/ocean polylines (shapefile already available on Datarmor; the reader also accepts the native GSHHG binary format). It is densified to half the finest cell size and thinned to one point per cell at $4\,N_{\mathrm{side}}^{\mathrm{max}}$. Nearest-coast queries use buckets of HEALPix cells at $N_{\mathrm{side}}^{\mathrm{max}}/4$, searched by rings of neighbours. The number of levels is limited to $\log_2(N_{\mathrm{side}}^{\mathrm{max}}/N_{\mathrm{side}}^{\mathrm{min}}) \leq 4$.

2.2 2:1 balance

Leaves are then split until any two leaves sharing an edge *or a corner* differ by at most one level. A leaf at depth d is split when one of its eight same-depth neighbours is refined and one of that neighbour's children is refined as well.

2.3 Triangulation from the HEALPix adjacency

Every leaf registers the eight points of its boundary that it can share with a neighbour: its four corners and the midpoints of its four edges. The midpoints are the corners of its children. Identical

points coming from different leaves are merged by quantised unit vectors. Around a point p, the registered leaves sorted by angle in the tangent plane at p form a polygon of cell centres, the dual cell of p:

- 4 cells at a regular vertex;
- 3 cells at the eight base-face corners where only three cells meet;
- 3 cells at the midpoint of a coarse edge facing two fine cells.

Each maximal run of consecutive wet leaves with at least three members is triangulated by ear clipping, at each step clipping the ear with the largest minimum angle.

The mesh is conforming because two leaves sharing a boundary segment $[p, q]$ are consecutive around p and around q only. Their connecting edge therefore belongs to exactly the two polygons of p and q, and to at most one triangle in each. With the 2:1 balance, a quarter point of a coarse edge is registered by at most two fine leaves and never produces a triangle.

2.4 Ellipsoid

HEALPix is defined on a sphere. To use the WGS84 ellipsoid, we use the authalic sphere, which keeps HEALPix equal-area on the ellipsoid. The geodetic latitudes of all inputs (bathymetry, coastline, domain) are converted to authalic latitudes before hashing, with the exact formula. Node latitudes are converted back with a series in e^2 refined by two Newton steps; the round-trip error is below 10^{-12} rad. The difference between the two latitudes reaches 0.128° at 45°, i.e. more than 14 km, which is larger than the finest cells. A spherical option is kept for comparison.

2.5 Bathymetry and wet cells

Each GEBCO 2024 sample (15″) is assigned to the leaf containing it. A leaf is wet when at least half of its samples are wet and its mean wet depth exceeds 5 m. The node depth is the mean depth over the wet samples. Only the largest connected wet component is kept, which removes lakes and basins not connected to the ocean at the grid resolution.

3 Validation on synthetic cases

The mesh is checked automatically: no edge shared by more than two triangles, no interior edge traversed twice in the same direction, all triangles counter-clockwise, Euler characteristic $V - E + F = 1 - h$ for h holes, and the largest level jump across an edge. Table 1 gives the multi-resolution test: an island of radius 2° centred on a HEALPix base-face corner (45°E, 41.8°N), with $N_{\text{side}} = 32 \dots 512$ and rules (30 km, 512), (100 km, 256), (250 km, 128). Figure 1 shows the mesh around the island.

Quantity	Value
nodes (N_{side} 32/64/128/256/512)	94 / 194 / 231 / 297 / 494
triangles	2471
cells split by the 2:1 balance	29
non-manifold / inconsistent edges, clockwise triangles, pinch nodes	0 / 0 / 0 / 0
Euler characteristic (one island)	0
max level jump across an edge	1
min / max angle	17.0° / 143.7°
triangles with an angle $< 20°$	1

Table 1: Synthetic island test (`cargo test island_multiscale`).

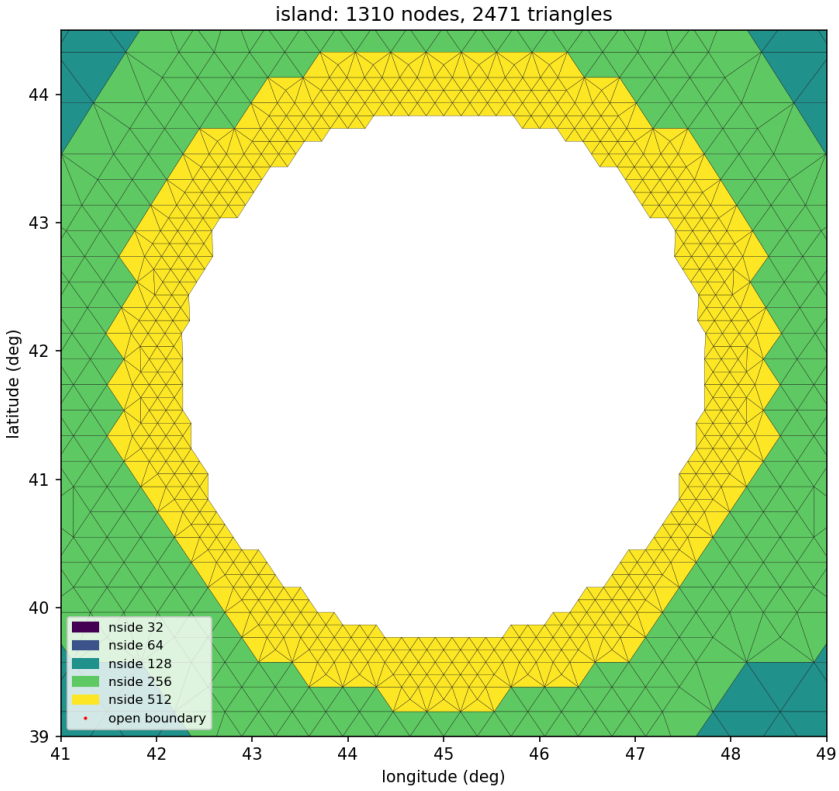


Figure 1: Multi-resolution HEALPix mesh around the synthetic island, coloured by N_{side}. The island sits on a base-face corner, where only three cells meet.

4 North Atlantic configuration

- Domain: 100°W–45°E, 0°–80°N, including the Mediterranean and the North Sea.
- Resolution: $N_{\mathrm{side}}^{\mathrm{min}} = 128$ (≈51 km) to $N_{\mathrm{side}}^{\mathrm{max}} = 2048$ (≈3.2 km).
- Refinement rules: 10 km → 2048, 50 km → 1024, 200 km → 512.
- Physics and numerics: WW3 `develop` (NOAA-EMC), ST4 source terms, 36 directions × 32 frequencies ($f_1 = 0.0373$ Hz, ratio 1.1), implicit unstructured scheme (PDLIB) with domain decomposition.
- Forcing (January 2026): ERA5 10 m wind; currents and sea level from Copernicus Marine; a fixed sea-ice obstruction taken from the January ERA5 sea-ice concentration.
- Open boundaries (southern boundary at the equator, eastern Barents Sea): spectra from an existing global run. [TODO: source of the spectra]

4.1 Grid statistics

The grid was built on Datarmor (PBS job 7893701, 8 cores) in 152 s wall clock (a first build, job 7893385, took 375 s with a peak resident memory of 5.8 GB); the GEBCO extraction (36 240 × 20 640 samples at 15 arcsec) takes 30 s. Coastline: GSHHS full-resolution land polylines, densified to 883 611 points. Table 3 summarises the mesh. The largest wet component is kept (902 components found); the Euler characteristic −693 corresponds to 694 islands.

Pinch nodes. WW3 walks the boundary around every node and stops (`w3triamd`: "dramatic inconsistency") when two triangle fans touch at a single node, a bow-tie that appears where water lies on two sides of a narrow spit. Such nodes pass the edge-based checks. The generator now keeps, at every such node, the largest fan and drops the triangles of the others, iterating to a fixed point: 671 triangles (0.07 %) were removed, and the validator counts pinch nodes (0 after the fix).

Triangle quality. The nodes are fixed HEALPix centres, so the angles can only be improved through the connectivity. The generator now (i) flips interior edges with the max-min angle criterion (Lawson's algorithm on the sphere, which converges to the Delaunay triangulation of the nodes constrained by the boundary), keeping a level jump of at most 1 across every edge, and (ii) removes boundary triangles whose angles are outside the accepted range, iterating and removing the pinch nodes this creates; interior triangles are never removed, so no hole appears. With `quality.enforce` the program fails if an angle is still out of range. On the North Atlantic (`configs/natl_v2_datarmor.json`, PBS job 7894086, 130 s, of which 8 s for the quality step), 14 750 flips and 1 275 removed boundary triangles bring the angles from 14.1°–139.7° (raw triangulation) to 20.7°–127.7° (Table 2). Three interior triangles keep an angle between 120° and 127.7°: no flip improves them (non-convex quadrilateral or no gain), and allowing a level jump of 2 across flipped edges gives the same mesh. Lowering them further would require moving nodes away from the pixel centres; the default range is therefore 20°–130°.

	v1 (no quality step)	v2 (flips + trim)
nodes / triangles	502 690 / 949 199	502 408 / 947 850
nodes at $N_{\mathrm{side}} = 2048$	277 607	277 325
edge flips / boundary triangles removed	–	14 750 / 1 275
min / max angle	17.5° / 139.7°	20.7° / 127.7°
triangles $< 20°$ / $> 120°$	3 781 / 3 070	0 / 3
Euler characteristic	−693	−682
defects (non-manifold, inconsistent, clockwise, pinch)	0	0
open-boundary nodes / max level jump	971 / 1	971 / 1

Table 2: Effect of the triangle-quality step on the North Atlantic grid. The January 2026 runs below use v1.

Open-boundary nodes (971) lie on the equator, at 80°N, on the 45°E cut of the Barents Sea and on the 100°W cut of the Canadian archipelago (Figure 2).

Quantity	Valu
nodes (N_{side} 128/256/512/1024/2048)	10 160 / 6 274 / 73 088 / 135 561 / 277 60
nodes, total	502 69
triangles	949 19
leaves of the quadtree (wet)	1 168 555 (573 61
cells split by the 2:1 balance	2 45
non-manifold / inconsistent edges, clockwise triangles, pinch nodes	0 / 0 / 0 /
Euler characteristic	−69
max level jump across an edge	
min / max angle	17.5° / 139.
triangles with an angle $< 20°$ / $> 120°$	3 781 / 3 07
open-boundary nodes	97
time: refinement + balance / cell bathymetry / triangulation	70 / 23 / 11

Table 3: North Atlantic grid (`configs/natl_datarmor.json`).

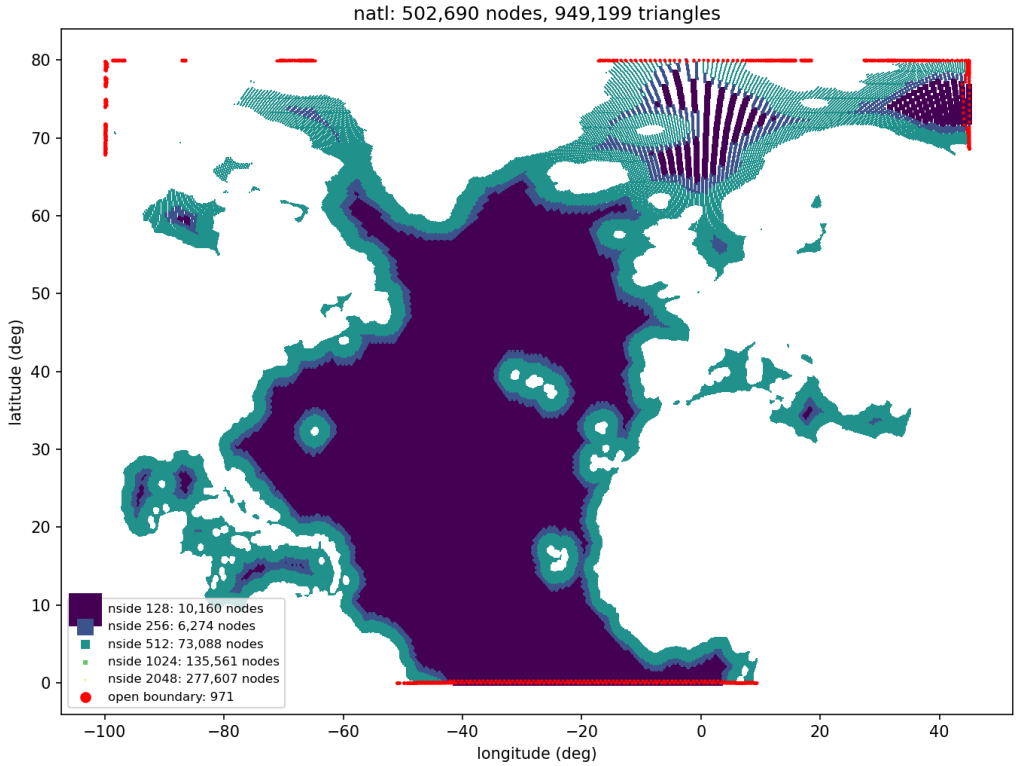


Figure 2: North Atlantic grid nodes coloured by N_{side}; open-boundary nodes in red.

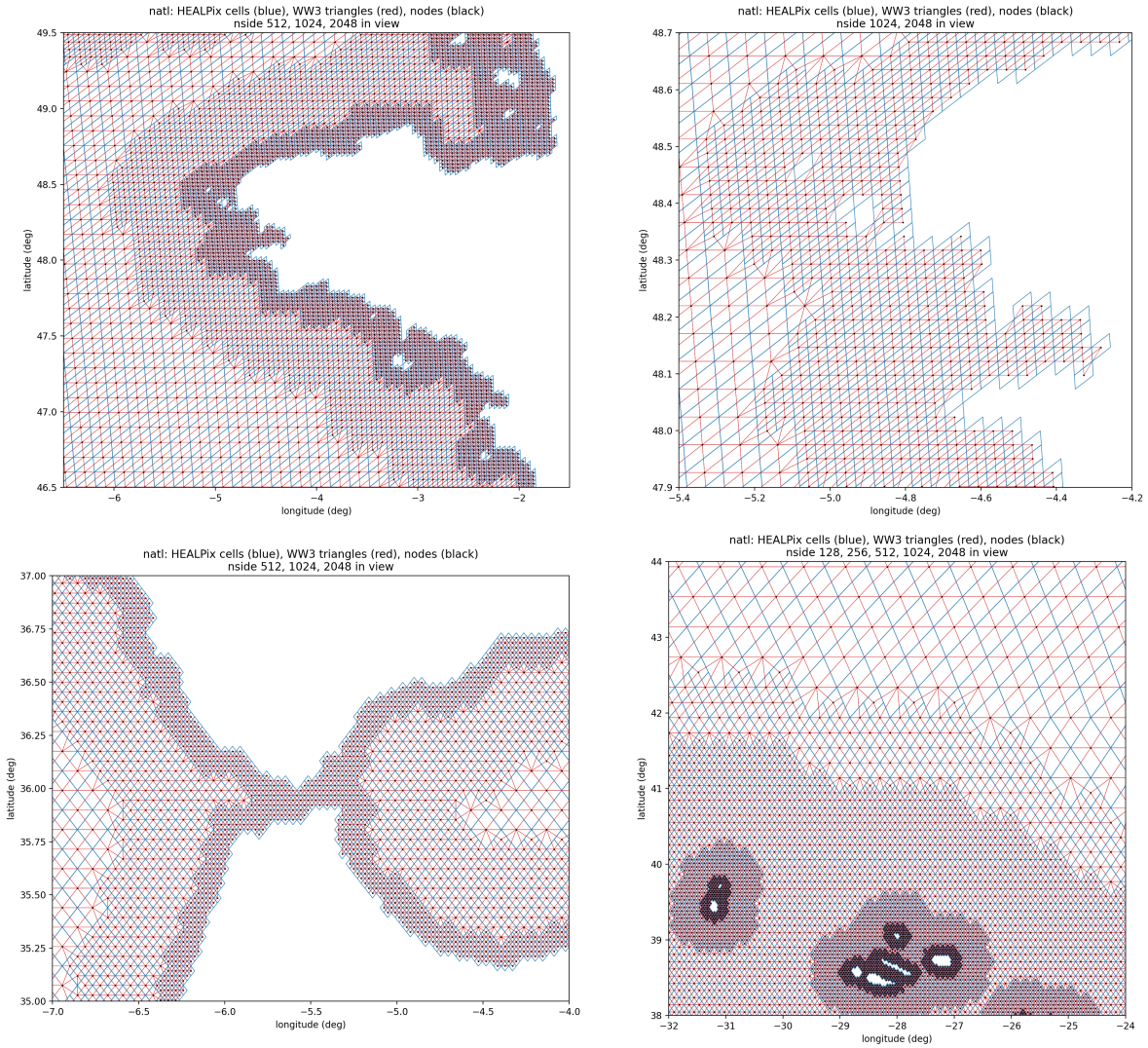


Figure 3: HEALPix cells (blue) and WW3 triangles (red) around Brittany, the Iroise Sea, the Strait of Gibraltar and the Azores (open ocean down to $N_{\mathrm{side}} = 128$). Near 0° longitude above 41.8°N the cells are strongly sheared in longitude–latitude because they lie at the boundary of two polar base faces.

4.2 WW3 build and grid preprocessing

WW3 `develop` (commit `dea641c`) is built on Datarmor with Intel Fortran 19.1.3 and Intel MPI 2019.9 (`scripts/build_ww3.pbs`, 5.5 min), with the switches of `ww3/switch_hww3`: PDLIB with METIS domain decomposition, ST4, DIST/MPI, currents and wind input, IC0/IS0 (ice as an obstruction only). Two local dependencies are built because the system ones do not fit: netCDF-Fortran 4.6.1 (the system Fortran module was made by an old gfortran) on top of the system netCDF-C 4.8.1, and ParMETIS 4.0.3 with METIS 5.1.0. Intel's preprocessor drops the quotes of `-DENDIANNESS='big_endian'`, so `FILE_ENDIAN` is set in `constants.F90` before compiling.

The open-boundary file follows `READMSHOBC`: a comment line, then `index 0 node` lines, ended by a comment line; a bare node list is silently read as empty. Depths are written positive in the Gmsh file and turned negative by `UNST%SF = -1`. `ww3_grid` (PBS job 7893702, 35 s) then reads 502 690 sea points and 971 active boundary points and writes `mod_def.ww3` (260 MB).

4.3 First run: homogeneous wind

Before the forcing is available, `ww3_shel` is run for 12 h from a cold start (1 January 2026, 00–12 UTC) with a homogeneous 15 m s^{-1} westerly wind, no current, water level or ice, and no boundary spectra (`ww3/natl_test`, `scripts/natl_test.pbs`, PBS job 7893925). The run uses 56 MPI ranks on two Datarmor nodes, with the ParMETIS decomposition and the fully implicit scheme of Table 3 at $\Delta t = 600$ s.

The 72 time steps take 1337 s, i.e. 18.6 s per step and a speed-up of 32 over real time; the Jacobi–Gauss–Seidel solver stops after about 20 iterations per step, when 1.1 % of the nodes are not yet converged (threshold `JGS_PMIN` = 3 %). Memory use is 55 GB over the two nodes, the restart file is 2.3 GB and `ww3_ounf` writes the hourly fields in 5 s. A full month (4464 steps) would therefore cost about 23 h on 56 cores, or 1300 core-hours.

The wave field grows as expected (Fig. 4): the median H_s is 1.0 m after 3 h, 1.85 m after 6 h and 2.75 m after 12 h, with T_{02} rising from 2.6 to 4.8 s. H_s stays lower on the lee side of the land masses (east of Britain, Shetland, the Americas) and at the open boundaries, where no spectra are imposed. The highest values (3.9 m) are on the Norwegian shelf near 61°N, where the waves shoal. No NaN appears.

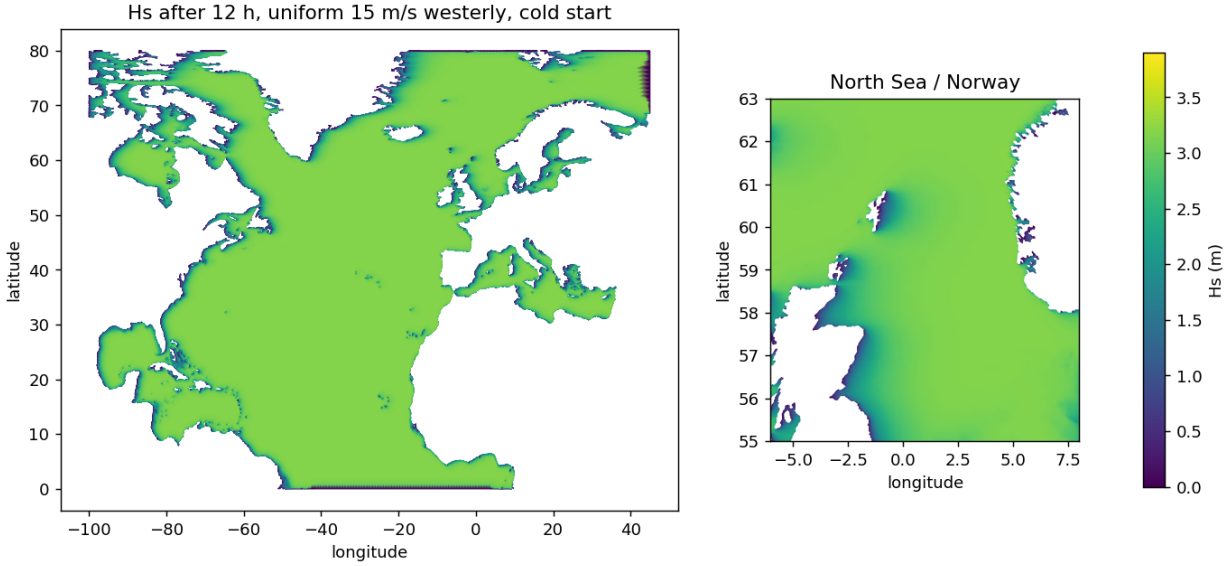


Figure 4: H_s after 12 h of a homogeneous 15 m s^{-1} westerly wind from rest (PBS job 7893925).

4.4 January 2026: forcing and runs

Forcing. ERA5 hourly 10 m wind and sea-ice cover (2°S–82°N, 102°W–47°E, 0.25°) are downloaded with `cdsapi` in 7 min (`scripts/get_era5.py`) and rewritten for `ww3_prnc` (`scripts/prep_era5.py`: ascending latitudes, CF time, `_FillValue`); the ice is the January mean, repeated at the first and last hour so that the obstruction is fixed. `ww3_prnc` interpolates the wind on the nodes in 72 s (`wind.ww3`, 3.0 GB; PBS job 7893994). Surface currents and sea level come from Copernicus Marine `GLOBAL_ANALYSISFORECAST_PHY_001_024` (hourly, 1/12°), read from the Datarmor mirror (`/home/ref-ocean-model-public/.../cmems_mod_glo_phy_anfc_0.083deg_PT1H-m`), taking for each day the latest forecast run (`scripts/prep_cmems.py`, PBS job 7894068, 14 min: `current.ww3` 3.0 GB, `level.ww3` 1.5 GB). Two pitfalls: the serial WW3 tools are MPI builds and need `I_MPI_FABRICS=shm` on a one-core node, and `ww3_prnc` stops if a field has no `_FillValue`.

Runs. Two runs cover 1 January 00 UTC to 31 January 23 UTC (last ERA5 hour) from rest on grid v1, with 112 MPI ranks on 4 nodes: wind and ice only (PBS job 7894000) and wind, ice,

currents and sea level (PBS job 7894069). The wind-and-ice run took 40 754 s (11.3 h, 9.1 s per 600 s step, 1270 core-hours) and `ww3_ounf` wrote the 744 hourly fields (6.1 GB). No boundary spectra are imposed yet. These two runs are kept on grid v1 as the reference January simulation; grid v2 (with the triangle-quality step, Table 2) is used for all later runs.

4.5 January 2026: validation against altimetry

The model H_s is compared with the along-track L3 NRT significant wave height of the ten altimeters of CMEMS `WAVE_GLO_PHY_SWH_L3_NRT_014_001` (AltiKa, CryoSat-2, CFOSAT, HY-2B, HY-2C, Jason-3, Sentinel-3A/B, Sentinel-6A, SWOT nadir; 4103 files, filtered and bias-corrected `VAVH`) with `scripts/validate_altimetry.py` (PBS job 7897833, 2 min). Each 1 Hz sample is matched to the nearest wet, ice-free node if that node lies within 0.75 of its HEALPix pixel size ($\sqrt{\pi/3}\,R/N_{\mathrm{side}}$: 51 km at $N_{\mathrm{side}} = 128$, 3.2 km at 2048), the model is interpolated linearly in time between hourly fields, the first three days (cold start) are skipped, and matched samples are averaged in blocks of 7 (about 50 km). There is no interpolation in space: the model value is the value of the cell that contains the sample, which is how a HEALPix field is meant to be read.

Over 4–31 January, 331 154 blocks give a bias of −0.19 m for a mean observed H_s of 2.92 m (−6.5 %), an RMSE of 0.45 m, a scatter index of 0.14 and a correlation of 0.972 (Table 4, Fig. 5). The ten missions agree within 0.08 m in bias. The error does not grow where the grid is coarse: bias and RMSE decrease from $N_{\mathrm{side}} = 128$ (open ocean, high seas) to 2048 (coast, lower seas), and the scatter index stays between 0.13 and 0.17. The along-track differences (Fig. 6) point to two causes of the negative bias, both outside the grid: (i) the southern open boundary at the equator, where no spectra are imposed, so that the swell coming from the South Atlantic is missing (differences below −1 m along 0°–5°N); (ii) the North-West Atlantic winter storms (40°–65°N, 30°–60°W), underestimated by ERA5 winds with the default ST4 growth parameter (`BETAMAX`=1.43, tuned for operational ECMWF winds; ERA5 usually needs about 1.6–1.75). The daily bias is −0.34 m on day 4 (swell still developing after the cold start), −0.13 m on days 8–11 and −0.25 m to −0.31 m during the storms of 26–31 January.

subset	N	mean obs (m)	bias (m)	RMSE (m)	SI / r
all	331 154	2.92	−0.19	0.45	0.14 / 0.972
$N_{\mathrm{side}} = 128$	183 701	3.20	−0.20	0.47	0.13 / 0.972
$N_{\mathrm{side}} = 256$	31 675	2.99	−0.20	0.46	0.14 / 0.970
$N_{\mathrm{side}} = 512$	82 837	2.65	−0.18	0.43	0.15 / 0.970
$N_{\mathrm{side}} = 1024$	29 615	2.05	−0.14	0.36	0.16 / 0.968
$N_{\mathrm{side}} = 2048$	3 326	1.82	−0.12	0.33	0.17 / 0.967
North Sea	5 567	2.90	−0.31	0.53	0.15 / 0.978
Mediterranean	14 539	2.19	−0.20	0.40	0.16 / 0.967

Table 4: Run 7894000 (grid v1, ERA5 wind and ice) against L3 altimetry, 4–31 January 2026, 50 km blocks; level = N_{side} of the matched node.

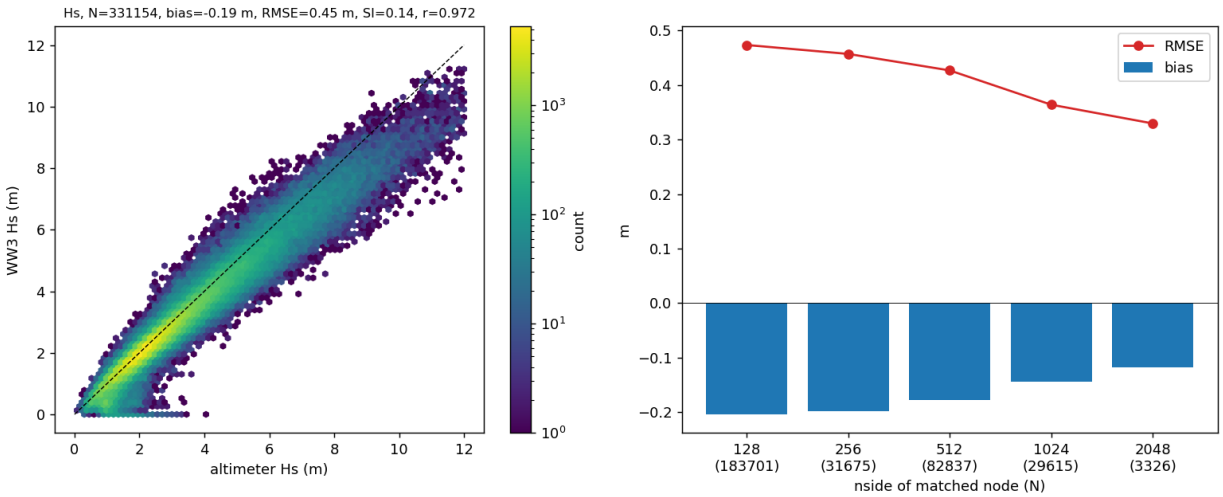


Figure 5: Left: WW3 against altimeter H_s (density, log scale). Right: bias and RMSE by N_{side} of the matched node (number of blocks in brackets).

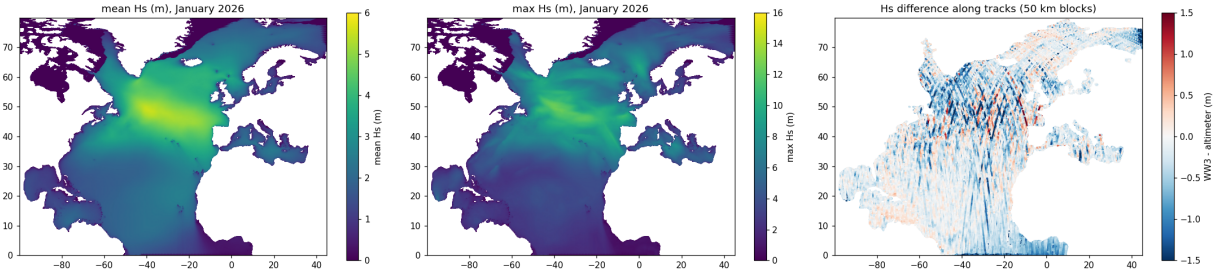


Figure 6: January 2026, run 7894000: mean and maximum H_s, and WW3 minus altimeter H_s along the tracks.

Effect of currents and sea level. The second run (PBS job 7894069, same forcing plus Copernicus Marine hourly surface currents and sea level, 41 654 s, 2 % slower) is validated in the same way (PBS job 7898389) on the same 331 154 blocks (Table 5). The currents lower the random error at every level (scatter index 0.139 to 0.138, correlation 0.9724 to 0.9738), as expected from wave refraction by currents, but make the mean bias 0.035 m more negative. WW3 forces the waves with the wind relative to the surface current, and in the North Atlantic the mean surface current mostly runs with the westerlies, which lowers the effective wind. This bias shift is of the order of the growth-parameter question above and should be settled together with it.

subset	bias wind / +cur (m)	RMSE wind / +cur (m)	SI wind / +cur	r wind / +cur
all	−0.191 / −0.225	0.450 / 0.462	0.139 / 0.138	0.9724 / 0.9738
$N_{\text{side}} = 128$	−0.204 / −0.240	0.473 / 0.486	0.133 / 0.132	0.9718 / 0.9730
$N_{\text{side}} = 512$	−0.178 / −0.212	0.427 / 0.437	0.146 / 0.145	0.9702 / 0.9717
$N_{\text{side}} = 2048$	−0.118 / −0.143	0.329 / 0.337	0.169 / 0.168	0.9668 / 0.9679
North Sea	−0.308 / −0.347	0.526 / 0.557	0.147 / 0.150	0.9780 / 0.9787
Mediterranean	−0.202 / −0.225	0.397 / 0.405	0.156 / 0.154	0.9672 / 0.9689

Table 5: Runs 7894000 (wind and ice) and 7894069 (plus currents and sea level) against L3 altimetry, 4–31 January 2026, same blocks.

4.6 HEALPix pyramid output

Because every node is a HEALPix pixel, the hourly H_s of the run is written as a Zarr V3 pyramid (`scripts/hs_pyramid.py`), one level per depth from 7 to 11: at a depth finer than a node its value is copied to all its descendants, at a coarser depth a pixel is the mean of its four children that hold data. The finest level is thus the model field cell by cell and each coarser level its exact area average. The store follows the GRID4EARTH production layout (`multiscales` and `dggs` conventions, nested `cell_ids` on the WGS84 ellipsoid, sharded zstd chunks). For January (744 hours) it holds 4 991 755 cells at $N_{\text{side}} = 2048$ and 20 991 at 128, costs 9 min on one core and 2.5 GB (1.6 times the raw node field), and is read without credentials by the gridlook preview, which picks the level from the zoom: `https://grid4earth.eu/gridlook-preview/#https://data.grid4earth.eu/trials/jmdelouis-2026-11-12-hww3/natl_jan2026/hs-l11-l7.zarr` (run with currents: same path with `natl_jan2026_cur`).

A third run on grid v2 with `BETAMAX` raised from 1.43 to 1.75 and the same forcing as the currents run (PBS jobs 7899025 for `ww3_grid` and the interpolations, 7899026 for the month) is under way.

[TODO: results of the v2 / BETAMAX run, boundary spectra from a global run]

A Reproducing

The complete procedure (sources and data, grid, WW3 build, forcing, runs, figures) is described step by step in the Sphinx documentation, `docs/reproducing.md`, built by the CI with the Rust API reference.

- Table 2: `qsub -v CFG=configs/natl_v2_datarmor.json,GRID=grids/natl_v2 scripts/natl_grid.pb`
- Table 1: `cd hww3-grid && cargo test -release island_multiscale - -nocapture`.
- Figure 1: `python scripts/plot_mesh.py /tmp/hww3_out_island tex/hww3/fig/island_mesh_zoom.p 41 49 39 44.5`.
- Table 3, Figures 2–3: `qsub scripts/natl_grid.pbs` on Datarmor, then `python scripts/plot_levels.py grids/natl/plot/nodes.npz fig/natl_levels.png` and `python scripts/plot_cells_tris.py grids/natl/plot/<zone>.npz fig/natl_<zone>.png`.
- Figure 4: `qsub scripts/natl_test.pbs` on Datarmor, then `python scripts/plot_hs.py runs/natl_test/ww3.202601.nc fig/natl_test_hs.png`.